Lecture 6: Limit of Functions and Proving Limits Systematically

Mathematical Analysis

1 Limit of Functions

Definition 1.1 (Existence of a Limit at a Point). [def:limit-of-functions] Under metric space (A, d_A) and (B, d_B), let $f : A \to B$ be a function. Let c be a limit point of A.
If there exists a point $\ell \in B$ such that

$$\forall \varepsilon \in \mathbb{R}_+,\ \exists \delta \in \mathbb{R}_+,\ \forall x \in A,\ 0 < d_A(x, c) < \delta \implies d_B(f(x), \ell) < \varepsilon$$

then we say that f has a limit at c

Theorem 1.1 (Uniqueness of a Function Limit). [thm:function-limit-uniqueness] *If a function f has a limit at a point c, then the limit is unique, ie*

$$\exists \ell : B,\ \forall \varepsilon \in \mathbb{R}_+,\ \exists \delta \in \mathbb{R}_+,\ \forall x \in A,\ 0 < d_A(x, c) < \delta \implies d_B(f(x), \ell) < \varepsilon$$

$$\iff$$

$$\exists! \ell : B,\ \forall \varepsilon \in \mathbb{R}_+,\ \exists \delta \in \mathbb{R}_+,\ \forall x \in A,\ 0 < d_A(x, c) < \delta \implies d_B(f(x), \ell) < \varepsilon$$

Proof. **TODO:** Prove uniqueness using the limit-point assumption and the triangle inequality □

Remark 1.1 (Board Comparison). [rem:sequence-function-limit-challenges] For a sequence $a : \mathbb{N}_1 \to X$ in a metric space (X, d) and a function $f : A \to B$, compare the two challenges

/	**Sequence**	**Function**
Arbitrary tolerance	$\forall \varepsilon \in \mathbb{R}_+$	$\forall \varepsilon \in \mathbb{R}_+$
Restrictor	$\exists N \in \mathbb{N}_1$	$\exists \delta \in \mathbb{R}_+$
Input restriction	$n \geq N$	$0 < d_A(x, c) < \delta$
Output requirement	$d(a(n), \ell) < \varepsilon$	$d_B(f(x), \ell) < \varepsilon$

The output requirement must hold for every $n \in \mathbb{N}_1$ or $x \in A$ satisfying its input restriction

1. ε is arbitrary small
2. δ and N is used to restrict x st x is close enough to the limit
3. δ and N are all depends on ε

Convention 1.1 (The ε–δ Challenge). [conv:epsilon-delta-challenge] A value $\ell \in B$ passes the ε–δ *challenge* for f at c if, for any proposed $\varepsilon > 0$, there exists a radius $\delta > 0$ such that for all $x \in A$ with $0 < d_A(x, c) < \delta$, $d_B(f(x), \ell) < \varepsilon$

Theorem 1.2 (Function Limit Challenge Replacement). [thm:function-limit-challenge-replacement] *Let (A, d_A) and (B, d_B) be metric spaces, let $f : A \to B$, let $c \in A$ be a limit point of A, and fix $\ell \in B$. Let g be a convergence challenge function. With balls taken in (A, d_A),*

$$\big(\forall \varepsilon \in \mathbb{R}_+,\ \exists \delta \in \mathbb{R}_+,\ \forall x \in B_\delta(c) \setminus \{c\},\quad d_B(f(x), \ell) < \varepsilon\big)$$

$$\iff$$

$$\big(\forall \varepsilon \in \mathbb{R}_+,\ \exists \delta \in \mathbb{R}_+,\ \forall x \in B_\delta(c) \setminus \{c\},\quad d_B(f(x), \ell) < g(\varepsilon)\big)$$

Proof. **TODO:** Prove both directions using positivity and arbitrarily small values of the challenge function □

Corollary 1.1 (Error-Bound Criterion). [cor:function-limit-error-bound] *Let (A, d_A) and (B, d_B) be metric spaces, let $f : A \to B$, let $c \in A$ be a limit point of A, and fix $\ell \in B$. Let g be a convergence challenge function, and let $\rho : \mathbb{R}_+ \to \mathbb{R}_+$. If, with balls taken in (A, d_A),*

$$\forall t \in \mathbb{R}_+, \quad \forall x \in B_{\rho(t)}(c) \setminus \{c\}, \quad d_B(f(x), \ell) \leq g(t)$$

then ℓ passes the ε–δ challenge for f at c

Proof. **TODO:** Choose a parameter with a sufficiently small error bound, then use its positive radius □

Corollary 1.2 (Sequence Error-Bound Criterion). [cor:sequence-limit-error-bound] ***TODO:*** *State the sequence criterion using a natural-number threshold M(t) and a challenge-function error bound g(t)*

Proof. **TODO:** Prove the sequence criterion by choosing a sufficiently small error bound and using the corresponding threshold □

Remark 1.2 (Using the Ball Radius as the Parameter). [rem:ball-radius-parameter] In the error-bound criterion, taking $\rho(t) = t$ lets us write $B_t(c)$ directly. Naming this parameter ε gives the sufficient condition

$$\forall \varepsilon \in \mathbb{R}_+, \quad \forall x \in B_\varepsilon(c) \setminus \{c\}, \quad d_B(f(x), \ell) \leq g(\varepsilon)$$

Here ε is the input-radius parameter, and $g(\varepsilon)$ is the output-error bound. A local restriction such as an input radius at most 1 can instead be included by taking $\rho(t) = \min(t, 1)$

Remark 1.3 (A Systematic Method for Proving Limits). [rem:systematic-limit-method] **TODO:** Summarize the shared procedure: choose a candidate limit, derive an error bound, verify the challenge function, and supply the corresponding radius or threshold

2 Function Limit Operator

Definition 2.1 (Limit of a Function at a Point). [def:function-limit-relation] **TODO:** Specify the ambient metric space and function domain, then define what it means for f to have limit ell at c

Definition 2.2 (Domain of the Function Limit Operator). [def:function-limit-domain] **TODO:** Define the restricted set of pairs (f,c), where c is a limit point of the domain and f has a limit at c

Definition 2.3 (Function Limit Operator). [def:function-limit-op] **TODO:** Give the operator's signature on the restricted pair domain and characterize its output using the limit relation

Convention 2.1 (Function Limit Notation). [conv:function-limit-notation] **TODO:** Relate the usual function-limit notation to the operator applied to (f,c)

Remark 2.1 (Well-Definedness of the Function Limit). [rem:function-limit-well-defined] **TODO:** Explain existence from the domain restriction and uniqueness from the uniqueness theorem

3 One-Sided Limits

Definition 3.1 (Right-Hand Limit). [def:right-hand-limit] **TODO:** Specify the real domain and right-hand accumulation assumption, then give the limit condition for inputs approaching c from the right

Definition 3.2 (Left-Hand Limit). [def:left-hand-limit] **TODO:** Specify the real domain and left-hand accumulation assumption, then give the limit condition for inputs approaching c from the left

Convention 3.1 (One-Sided Limit Notation). [conv:one-sided-limit-notation] **TODO:** Introduce the notation for right-hand and left-hand limits

Theorem 3.1 (Uniqueness of One-Sided Limits). [thm:one-sided-limit-uniqueness] ***TODO:*** *State uniqueness under the corresponding one-sided accumulation assumption*

Proof. **TODO:** Adapt the uniqueness argument to the one-sided input restriction □

Theorem 3.2 (Two-Sided and One-Sided Limits). [thm:two-sided-one-sided-limits] ***TODO:*** *State when a two-sided limit is equivalent to equal right-hand and left-hand limits, including the required domain assumptions*

Proof. **TODO:** Prove both directions □

Remark 3.1 (Error Bounds for One-Sided Limits). [rem:one-sided-error-bounds] **TODO:** Adapt the shared error-bound method to the right-hand and left-hand input regions

4 Limits at Infinity

Definition 4.1 (Limit at Positive Infinity). [def:limit-at-positive-infinity] **TODO:** Assume the real domain is unbounded above and define a finite output limit as the input tends to positive infinity

Definition 4.2 (Limit at Negative Infinity). [def:limit-at-negative-infinity] **TODO:** Assume the real domain is unbounded below and define a finite output limit as the input tends to negative infinity

Convention 4.1 (Limit Notation at Infinity). [conv:limit-at-infinity-notation] **TODO:** Introduce the notation for limits as the input tends to positive or negative infinity

Theorem 4.1 (Uniqueness of Limits at Infinity). [thm:limit-at-infinity-uniqueness] ***TODO:*** *State uniqueness under the corresponding unbounded-domain assumption*

Proof. **TODO:** Adapt the uniqueness argument using the corresponding input tail □

Remark 4.1 (Error Bounds for Limits at Infinity). [rem:infinity-error-bounds] **TODO:** Adapt the shared error-bound method using a real input threshold M(t)